
Electrodynamics

Study Notes: Theory, Examples & Problems

Vector Analysis • Electrostatics • Potentials
Electric Fields in Matter • Magnetostatics

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Chapter 1: Vector Analysis

Theory

Q1. Describe the dot and cross product of two vectors.

Dot product: $\mathbf{A} \cdot \mathbf{B} = AB \cos \theta$, where θ is the angle between the vectors when placed tail to tail. The result is a scalar.

Cross product: $\mathbf{A} \times \mathbf{B} = AB \sin \theta \hat{\mathbf{n}}$, where $\hat{\mathbf{n}}$ is a unit vector perpendicular to the plane of $\mathbf{A}$ and $\mathbf{B}$ (direction given by the right-hand rule).

Q2. Discuss the scalar triple product and the vector triple product.

Since the cross product of two vectors is itself a vector, it can be dotted or crossed with a third vector to form a triple product.

(i) **Scalar triple product:** $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$. Geometrically it is the volume of the parallelepiped generated by $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$: $|\mathbf{B} \times \mathbf{C}|$ is the area of the base and $|\mathbf{A} \cos \theta|$ is the altitude.

(ii) **Vector triple product:** $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$, simplified by the *BAC-CAB rule*:

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}).$$

Q3. Define the del operator.

$$\nabla = \hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z}$$

∇ is a vector operator: it is an instruction to differentiate whatever follows it.

Q4. Describe the gradient, divergence and curl.

Gradient (of a scalar T):

$$\nabla T = \frac{\partial T}{\partial x} \hat{\mathbf{x}} + \frac{\partial T}{\partial y} \hat{\mathbf{y}} + \frac{\partial T}{\partial z} \hat{\mathbf{z}}$$

∇T points in the direction of maximum increase of T ; its magnitude $|\nabla T|$ gives the slope (rate of increase) along that direction.

Divergence (of a vector $\mathbf{v}$):

$$\nabla \cdot \mathbf{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$$

It measures how much $\mathbf{v}$ spreads out (diverges) from the point in question.

Curl:

$$\nabla \times \mathbf{v} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_x & v_y & v_z \end{vmatrix} = \hat{\mathbf{x}} \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) + \hat{\mathbf{y}} \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) + \hat{\mathbf{z}} \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right)$$

It measures how much $\mathbf{v}$ swirls around the point in question.

Q5. Discuss the fundamental theorems for gradients, divergences and curls.

Gradient theorem: the line integral of a gradient is given by the value of the function at the end points:

$$\int_{\mathbf{a}}^{\mathbf{b}} (\nabla T) \cdot d\mathbf{l} = T(\mathbf{b}) - T(\mathbf{a})$$

Divergence (Gauss's) theorem: the integral of a divergence over a volume equals the flux through the bounding surface:

$$\int_V (\nabla \cdot \mathbf{v}) d\tau = \oint_S \mathbf{v} \cdot d\mathbf{a}$$

Curl (Stokes') theorem: the integral of a curl over a surface equals the circulation around its boundary:

$$\int_S (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = \oint_P \mathbf{v} \cdot d\mathbf{l}$$

Examples

Example 1.1 Let $\mathbf{C} = \mathbf{A} - \mathbf{B}$; calculate $\mathbf{C} \cdot \mathbf{C}$.

$$\mathbf{C} \cdot \mathbf{C} = (\mathbf{A} - \mathbf{B}) \cdot (\mathbf{A} - \mathbf{B}) = \mathbf{A} \cdot \mathbf{A} - 2\mathbf{A} \cdot \mathbf{B} + \mathbf{B} \cdot \mathbf{B} \Rightarrow C^2 = A^2 + B^2 - 2AB \cos \theta$$

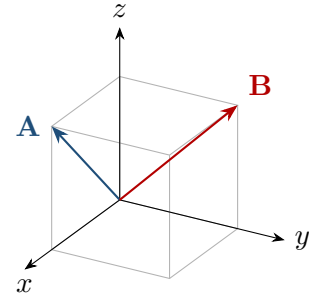
(the law of cosines).

Example 1.2 Find the angle between two face diagonals of a unit cube.

Take $\mathbf{A} = \hat{\mathbf{x}} + \hat{\mathbf{z}}$, $\mathbf{B} = \hat{\mathbf{y}} + \hat{\mathbf{z}}$. Then

$$\mathbf{A} \cdot \mathbf{B} = 1, \quad AB = \sqrt{2}\sqrt{2} = 2$$

$$\cos \theta = \frac{1}{2} \Rightarrow \theta = 60^\circ$$



Example 1.3 Find the gradient of $r = \sqrt{x^2 + y^2 + z^2}$.

$$\nabla r = \frac{x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}}{\sqrt{x^2 + y^2 + z^2}} = \frac{\mathbf{r}}{r} = \hat{\mathbf{r}}$$

Example 1.4 Divergences of $\mathbf{v}_a = \mathbf{r} = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}$, $\mathbf{v}_b = \hat{\mathbf{z}}$, $\mathbf{v}_c = z\hat{\mathbf{z}}$:

$$\nabla \cdot \mathbf{v}_a = 1 + 1 + 1 = 3, \quad \nabla \cdot \mathbf{v}_b = 0, \quad \nabla \cdot \mathbf{v}_c = 1.$$

Example 1.5 Curls of $\mathbf{v}_a = -y\hat{\mathbf{x}} + x\hat{\mathbf{y}}$ and $\mathbf{v}_b = x\hat{\mathbf{y}}$:

$$\nabla \times \mathbf{v}_a = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial_x & \partial_y & \partial_z \\ -y & x & 0 \end{vmatrix} = 2\hat{\mathbf{z}}, \quad \nabla \times \mathbf{v}_b = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial_x & \partial_y & \partial_z \\ 0 & x & 0 \end{vmatrix} = \hat{\mathbf{z}}.$$

Example 1.6 Line integral of $\mathbf{v} = y^2\hat{\mathbf{x}} + 2x(y+1)\hat{\mathbf{y}}$ from $\mathbf{a} = (1, 1, 0)$ to $\mathbf{b} = (2, 2, 0)$ along paths (1) and (2). What is $\oint \mathbf{v} \cdot d\mathbf{l}$ for the closed loop?

$d\mathbf{l} = dx\hat{\mathbf{x}} + dy\hat{\mathbf{y}} + dz\hat{\mathbf{z}}$.

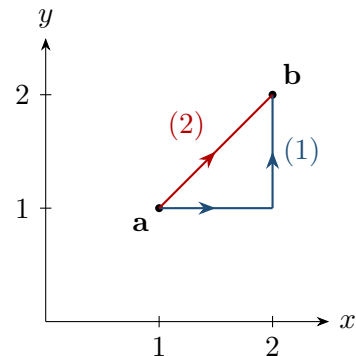
Path (1), horizontal part: $y = 1$, $d\mathbf{l} = dx\hat{\mathbf{x}}$, $\mathbf{v} \cdot d\mathbf{l} = y^2 dx = dx$,

$$\int \mathbf{v} \cdot d\mathbf{l} = \int_1^2 dx = 1.$$

Vertical part: $x = 2$, $d\mathbf{l} = dy\hat{\mathbf{y}}$, $\mathbf{v} \cdot d\mathbf{l} = 4(y+1) dy$,

$$\int \mathbf{v} \cdot d\mathbf{l} = 4 \int_1^2 (y+1) dy = 10.$$

So along path (1): $\int_a^b \mathbf{v} \cdot d\mathbf{l} = 1 + 10 = 11$.



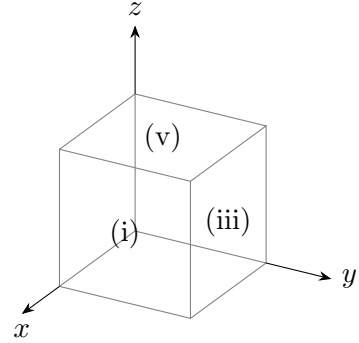
Path (2): $y = x$, $dy = dx$, $\mathbf{v} \cdot d\mathbf{l} = x^2 dx + 2x(x+1)dx = (3x^2 + 2x) dx$,

$$\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{v} \cdot d\mathbf{l} = \int_1^2 (3x^2 + 2x) dx = (x^3 + x^2) \Big|_1^2 = 10.$$

For the loop out along (1) and back along (2): $\oint \mathbf{v} \cdot d\mathbf{l} = 11 - 10 = 1$.

Example 1.7 Surface integral of $\mathbf{v} = 2xz\hat{\mathbf{x}} + (x+2)\hat{\mathbf{y}} + y(z^2-3)\hat{\mathbf{z}}$ over five sides (excluding the bottom) of the cube of side 2; outward is positive.

- (i) $x = 2$: $d\mathbf{a} = dy dz \hat{\mathbf{x}}$, $\mathbf{v} \cdot d\mathbf{a} = 4z dy dz$;
 $\int = 4 \int_0^2 dy \int_0^2 z dz = 16$
- (ii) $x = 0$: $d\mathbf{a} = -dy dz \hat{\mathbf{x}}$, $\mathbf{v} \cdot d\mathbf{a} = 0$; $\int = 0$
- (iii) $y = 2$: $d\mathbf{a} = dx dz \hat{\mathbf{y}}$; $\int = \int_0^2 (x+2) dx \int_0^2 dz = 12$
- (iv) $y = 0$: $d\mathbf{a} = -dx dz \hat{\mathbf{y}}$; $\int = -12$
- (v) $z = 2$: $d\mathbf{a} = dx dy \hat{\mathbf{z}}$, $\mathbf{v} \cdot d\mathbf{a} = y dx dy$;
 $\int = \int_0^2 dx \int_0^2 y dy = 4$



$$\int_{\text{surface}} \mathbf{v} \cdot d\mathbf{a} = 16 + 0 + 12 - 12 + 4 = 20$$

Example 1.8 Volume integral of $T = xyz^2$ over the prism ($0 \leq z \leq 3$, triangular base $x + y \leq 1$ in the first quadrant):

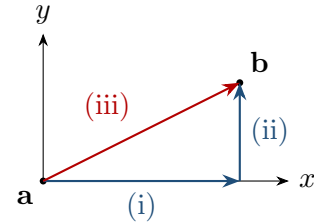
$$\int T d\tau = \int_0^3 z^2 \left\{ \int_0^1 y \left[\int_0^{1-y} x dx \right] dy \right\} dz = \frac{1}{2} \int_0^3 z^2 dz \int_0^1 (1-y)^2 y dy = \frac{1}{2} \cdot 9 \cdot \frac{1}{12} = \frac{3}{8}$$

Example 1.9 Check the gradient theorem for $T = xy^2$, $\mathbf{a} = (0, 0, 0)$, $\mathbf{b} = (2, 1, 0)$. Here $\nabla T = y^2\hat{\mathbf{x}} + 2xy\hat{\mathbf{y}}$ and $T(\mathbf{b}) - T(\mathbf{a}) = 2$.

Path (i)+(ii): along $y = 0$, $\nabla T \cdot d\mathbf{l} = y^2 dx = 0$. Then along $x = 2$, $\nabla T \cdot d\mathbf{l} = 4y dy$, $\int_0^1 4y dy = 2$. Total = 2 ✓

Path (iii): $y = x/2$, $dy = dx/2$,

$$\nabla T \cdot d\mathbf{l} = y^2 dx + 2xy dy = \frac{3}{4}x^2 dx, \quad \int_0^2 \frac{3}{4}x^2 dx = 2 \quad \checkmark$$



Example 1.10 Check the divergence theorem for $\mathbf{v} = y^2\hat{\mathbf{x}} + (2xy + z^2)\hat{\mathbf{y}} + (2yz)\hat{\mathbf{z}}$ on the unit cube at the origin.

$$\nabla \cdot \mathbf{v} = 2(x+y), \quad \int_V 2(x+y) d\tau = 2 \int_0^1 \int_0^1 \int_0^1 (x+y) dx dy dz = 2$$

Faces: (i) $x = 1$: $\int_0^1 \int_0^1 y^2 dy dz = \frac{1}{3}$; (ii) $x = 0$: $-\frac{1}{3}$; (iii) $y = 1$: $\iint (2x + z^2) dx dz = \frac{4}{3}$; (iv) $y = 0$: $-\iint z^2 dx dz = -\frac{1}{3}$; (v) $z = 1$: $\iint 2y dx dy = 1$; (vi) $z = 0$: 0.

$$\oint \mathbf{v} \cdot d\mathbf{a} = \frac{1}{3} - \frac{1}{3} + \frac{4}{3} - \frac{1}{3} + 1 + 0 = 2 \quad \checkmark$$

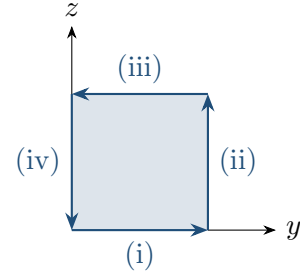
Example 1.11 Check Stokes' theorem for $\mathbf{v} = (2xz + 3y^2)\hat{\mathbf{y}} + (4yz^2)\hat{\mathbf{z}}$ on the unit square in the yz -plane.

$\nabla \times \mathbf{v} = (4z^2 - 2x)\hat{\mathbf{x}} + 2z\hat{\mathbf{z}}$, $d\mathbf{a} = dy dz \hat{\mathbf{x}}$. With $x = 0$:

$$\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = \int_0^1 \int_0^1 4z^2 dy dz = \frac{4}{3}$$

Line integral ($x = 0$): (i) $z = 0$: $\int_0^1 3y^2 dy = 1$; (ii) $y = 1$: $\int_0^1 4z^2 dz = \frac{4}{3}$; (iii) $z = 1$: $\int_1^0 3y^2 dy = -1$; (iv) $y = 0$: 0.

$$\oint \mathbf{v} \cdot d\mathbf{l} = 1 + \frac{4}{3} - 1 + 0 = \frac{4}{3} \checkmark$$



Example 1.12 Evaluate $\int_0^\infty x e^{-x} dx$ by parts, with $f = x$, $dg = e^{-x} dx$, $g = -e^{-x}$:

$$\int_0^\infty x e^{-x} dx = \int_0^\infty e^{-x} dx - x e^{-x} \Big|_0^\infty = -e^{-x} \Big|_0^\infty = 1$$

Example 1.13 Volume of a sphere of radius R :

$$V = \int_0^R \int_0^\pi \int_0^{2\pi} r^2 \sin \theta dr d\theta d\phi = \left(\frac{R^3}{3}\right)(2)(2\pi) = \frac{4}{3}\pi R^3$$

Example 1.14 $\int_0^3 x^3 \delta(x-2) dx = 2^3 = 8$ (the delta function picks out the value at $x = 2$).

Example 1.15 Show that $\delta(kx) = \frac{1}{|k|} \delta(x)$ for any nonzero constant k (in particular $\delta(-x) = \delta(x)$). For a test function $f(x)$, substitute $y = kx$, $dx = dy/k$. If $k > 0$ the limits stay the same; if $k < 0$ they flip, giving a minus sign:

$$\int_{-\infty}^{\infty} f(x) \delta(kx) dx = \pm \frac{1}{k} \int_{-\infty}^{\infty} f(y/k) \delta(y) dy = \frac{1}{|k|} f(0) = \int_{-\infty}^{\infty} f(x) \left[\frac{1}{|k|} \delta(x) \right] dx$$

Since the two expressions agree for every test function, $\delta(kx) = \frac{1}{|k|} \delta(x)$.

Example 1.16 Evaluate $J = \int_V (r^2 + 2) \nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) d\tau$ over a sphere of radius R at the origin. Using $\nabla \cdot (\hat{\mathbf{r}}/r^2) = 4\pi\delta^3(\mathbf{r})$:

$$J = \int_V (r^2 + 2) 4\pi\delta^3(\mathbf{r}) d\tau = 4\pi(0 + 2) = 8\pi$$

Problems

Problem 1.2 Is the cross product associative, $(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} \stackrel{?}{=} \mathbf{A} \times (\mathbf{B} \times \mathbf{C})$?

No. Counterexample: let $\mathbf{A} = \mathbf{B}$ and $\mathbf{C} \perp \mathbf{A}$. Then $\mathbf{B} \times \mathbf{C}$ points out of the page and $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$ has magnitude $ABC \neq 0$. But $\mathbf{A} \times \mathbf{B} = \mathbf{0}$, so $(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} = \mathbf{0} \neq \mathbf{A} \times (\mathbf{B} \times \mathbf{C})$.

Problem 1.3 Angle between two body diagonals of a cube.

$$\mathbf{A} = \hat{\mathbf{x}} + \hat{\mathbf{y}} - \hat{\mathbf{z}}, \quad \mathbf{B} = \hat{\mathbf{x}} + \hat{\mathbf{y}} + \hat{\mathbf{z}}, \quad A = B = \sqrt{3}, \quad \mathbf{A} \cdot \mathbf{B} = 1$$

$$\cos \theta = \frac{1}{3} \Rightarrow \theta = \cos^{-1}(1/3) \approx 70.5^\circ$$

Problem 1.5 Prove the BAC–CAB rule in components. With $\mathbf{B} \times \mathbf{C} = (B_y C_z - B_z C_y)\hat{\mathbf{x}} + (B_z C_x - B_x C_z)\hat{\mathbf{y}} + (B_x C_y - B_y C_x)\hat{\mathbf{z}}$, the x -component of $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$ is

$$A_y(B_x C_y - B_y C_x) - A_z(B_z C_x - B_x C_z) = A_y B_x C_y - A_y B_y C_x - A_z B_z C_x + A_z B_x C_z.$$

The x -component of $\mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$ is

$$B_x(A_x C_x + A_y C_y + A_z C_z) - C_x(A_x B_x + A_y B_y + A_z B_z) = A_y B_x C_y + A_z B_x C_z - A_y B_y C_x - A_z B_z C_x.$$

These are equal; the y and z components follow in the same way. ✓

Problem 1.7 Separation vector from source (2, 8, 7) to field point (4, 6, 8):

$$\mathbf{r} = (4 - 2)\hat{\mathbf{x}} + (6 - 8)\hat{\mathbf{y}} + (8 - 7)\hat{\mathbf{z}} = 2\hat{\mathbf{x}} - 2\hat{\mathbf{y}} + \hat{\mathbf{z}}, \quad r = \sqrt{4 + 4 + 1} = 3, \quad \hat{\mathbf{r}} = \frac{2}{3}\hat{\mathbf{x}} - \frac{2}{3}\hat{\mathbf{y}} + \frac{1}{3}\hat{\mathbf{z}}$$

Problem 1.10

- (a) Translation $(\bar{x}, \bar{y}, \bar{z}) = (x, y - a, z)$: components do not change, $\bar{A}_x = A_x$, $\bar{A}_y = A_y$, $\bar{A}_z = A_z$.
- (b) Inversion $(\bar{x}, \bar{y}, \bar{z}) = (-x, -y, -z)$: $\mathbf{A} \rightarrow -\mathbf{A}$, i.e. $\bar{A}_i = -A_i$.
- (c) Cross product under inversion: $\mathbf{A} \times \mathbf{B} \rightarrow (-\mathbf{A}) \times (-\mathbf{B}) = \mathbf{A} \times \mathbf{B}$ — no sign change, so it is a *pseudovector*. The cross product of two pseudovectors is a pseudovector too. Examples in mechanics: angular momentum and torque.
- (d) Scalar triple product under inversion: $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) \rightarrow (-\mathbf{A}) \cdot ((-\mathbf{B}) \times (-\mathbf{C})) = -\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$. It changes sign — a *pseudoscalar*.

Problem 1.11 Gradients:

- (a) $f = x^2 + y^3 + z^4$: $\nabla f = 2x\hat{\mathbf{x}} + 3y^2\hat{\mathbf{y}} + 4z^3\hat{\mathbf{z}}$
- (b) $f = x^2 y^3 z^4$: $\nabla f = 2xy^3z^4\hat{\mathbf{x}} + 3x^2y^2z^4\hat{\mathbf{y}} + 4x^2y^3z^3\hat{\mathbf{z}}$
- (c) $f = e^x \sin y \ln z$: $\nabla f = e^x \sin y \ln z \hat{\mathbf{x}} + e^x \cos y \ln z \hat{\mathbf{y}} + \frac{e^x \sin y}{z} \hat{\mathbf{z}}$

Problem 1.15 Divergences:

- (a) $\mathbf{v}_a = x^2\hat{\mathbf{x}} + 3xz^2\hat{\mathbf{y}} - 2xz\hat{\mathbf{z}}$: $\nabla \cdot \mathbf{v}_a = 2x + 0 - 2x = 0$
- (b) $\mathbf{v}_b = xy\hat{\mathbf{x}} + 2yz\hat{\mathbf{y}} + 3zx\hat{\mathbf{z}}$: $\nabla \cdot \mathbf{v}_b = y + 2z + 3x$
- (c) $\mathbf{v}_c = y^2\hat{\mathbf{x}} + (2xy + z^2)\hat{\mathbf{y}} + 2yz\hat{\mathbf{z}}$: $\nabla \cdot \mathbf{v}_c = 0 + 2x + 2y = 2(x + y)$

Problem 1.16 Sketch $\mathbf{v} = \hat{\mathbf{r}}/r^2$ and compute its divergence.

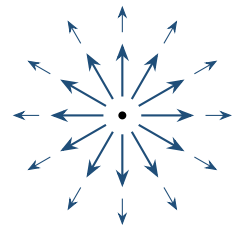
Write $\mathbf{v} = \frac{x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}}{(x^2 + y^2 + z^2)^{3/2}}$. Then

$$\frac{\partial}{\partial x} [x(x^2 + y^2 + z^2)^{-3/2}] = r^{-3} - 3x^2r^{-5},$$

and similarly for y, z , so

$$\nabla \cdot \mathbf{v} = 3r^{-3} - 3r^{-5}(x^2 + y^2 + z^2) = 3r^{-3} - 3r^{-3} = 0.$$

$\nabla \cdot \mathbf{v} = 0$ everywhere *except at the origin*, where the function blows up (this is the source of the $4\pi\delta^3(\mathbf{r})$).



Problem 1.18 Curls:

$$(a) \nabla \times \mathbf{v}_a = \hat{\mathbf{x}}(0 - 6xz) + \hat{\mathbf{y}}(0 + 2z) + \hat{\mathbf{z}}(3z^2 - 0) = -6xz \hat{\mathbf{x}} + 2z \hat{\mathbf{y}} + 3z^2 \hat{\mathbf{z}}$$

$$(b) \nabla \times \mathbf{v}_b = \hat{\mathbf{x}}(0 - 2y) + \hat{\mathbf{y}}(0 - 3z) + \hat{\mathbf{z}}(0 - x) = -2y \hat{\mathbf{x}} - 3z \hat{\mathbf{y}} - x \hat{\mathbf{z}}$$

$$(c) \nabla \times \mathbf{v}_c = \hat{\mathbf{x}}(2z - 2z) + \hat{\mathbf{y}}(0 - 0) + \hat{\mathbf{z}}(2y - 2y) = \mathbf{0}$$

Problem 1.25 Check product rules with $\mathbf{A} = x\hat{\mathbf{x}} + 2y\hat{\mathbf{y}} + 3z\hat{\mathbf{z}}$ and $\mathbf{B} = 3y\hat{\mathbf{x}} - 2x\hat{\mathbf{y}}$.

$$(a) \nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B}).$$

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ x & 2y & 3z \\ 3y & -2x & 0 \end{vmatrix} = 6xz \hat{\mathbf{x}} + 9zy \hat{\mathbf{y}} + (-2x^2 - 6y^2) \hat{\mathbf{z}}, \quad \nabla \cdot (\mathbf{A} \times \mathbf{B}) = 6z + 9z = 15z$$

$$\nabla \times \mathbf{A} = \mathbf{0} \Rightarrow \mathbf{B} \cdot (\nabla \times \mathbf{A}) = 0; \quad \nabla \times \mathbf{B} = -5\hat{\mathbf{z}} \Rightarrow \mathbf{A} \cdot (\nabla \times \mathbf{B}) = -15z. \text{ RHS} = 0 - (-15z) = 15z \checkmark$$

$$(b) \nabla(\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{A}.$$

$$\mathbf{A} \cdot \mathbf{B} = 3xy - 4xy = -xy, \quad \nabla(\mathbf{A} \cdot \mathbf{B}) = -y \hat{\mathbf{x}} - x \hat{\mathbf{y}}$$

$$\mathbf{A} \times (\nabla \times \mathbf{B}) = -10y \hat{\mathbf{x}} + 5x \hat{\mathbf{y}}; \quad \mathbf{B} \times (\nabla \times \mathbf{A}) = \mathbf{0}; \quad (\mathbf{A} \cdot \nabla)\mathbf{B} = 6y \hat{\mathbf{x}} - 2x \hat{\mathbf{y}}; \quad (\mathbf{B} \cdot \nabla)\mathbf{A} = 3y \hat{\mathbf{x}} - 4x \hat{\mathbf{y}}.$$

$$\text{Sum: } (-10y + 6y + 3y) \hat{\mathbf{x}} + (5x - 2x - 4x) \hat{\mathbf{y}} = -y \hat{\mathbf{x}} - x \hat{\mathbf{y}} \checkmark$$

$$(c) \nabla \times (\mathbf{A} \times \mathbf{B}) = (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B} + \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A}).$$

$$\nabla \times (\mathbf{A} \times \mathbf{B}) = \hat{\mathbf{x}}(-12y - 9y) + \hat{\mathbf{y}}(6x + 4x) + \hat{\mathbf{z}}(0) = -21y \hat{\mathbf{x}} + 10x \hat{\mathbf{y}}$$

$$\nabla \cdot \mathbf{A} = 6, \nabla \cdot \mathbf{B} = 0. \text{ RHS} = (3y\hat{\mathbf{x}} - 4x\hat{\mathbf{y}}) - (6y\hat{\mathbf{x}} - 2x\hat{\mathbf{y}}) + 0 - 6(3y\hat{\mathbf{x}} - 2x\hat{\mathbf{y}}) = -21y \hat{\mathbf{x}} + 10x \hat{\mathbf{y}} \checkmark$$

Problem 1.26 Laplacians:

$$(a) T_a = x^2 + 2xy + 3z + 4: \quad \nabla^2 T_a = 2$$

$$(b) T_b = \sin x \sin y \sin z: \text{ each second derivative gives } -T_b, \text{ so } \nabla^2 T_b = -3 \sin x \sin y \sin z$$

$$(c) T_c = e^{-5x} \sin 4y \cos 3z: \quad \nabla^2 T_c = (25 - 16 - 9)T_c = 0$$

$$(d) \mathbf{v} = x^2 \hat{\mathbf{x}} + 3xz^2 \hat{\mathbf{y}} - 2xz \hat{\mathbf{z}}: \quad \nabla^2 v_x = 2, \nabla^2 v_y = 6x, \nabla^2 v_z = 0, \text{ so } \nabla^2 \mathbf{v} = 2 \hat{\mathbf{x}} + 6x \hat{\mathbf{y}}$$

Problem 1.32 Check the gradient theorem for $T = x^2 + 4xy + 2yz^3$, $\mathbf{a} = (0, 0, 0)$, $\mathbf{b} = (1, 1, 1)$.

$$T(\mathbf{b}) - T(\mathbf{a}) = 1 + 4 + 2 - 0 = 7, \quad \nabla T = (2x + 4y)\hat{\mathbf{x}} + (4x + 2z^3)\hat{\mathbf{y}} + 6yz^2\hat{\mathbf{z}}$$

$$(a) (0, 0, 0) \rightarrow (1, 0, 0) \rightarrow (1, 1, 0) \rightarrow (1, 1, 1): \quad \int_0^1 2x dx = 1, \quad \int_0^1 4 dy = 4, \quad \int_0^1 6z^2 dz = 2. \text{ Total} = 7 \checkmark$$

$$(b) (0, 0, 0) \rightarrow (0, 0, 1) \rightarrow (0, 1, 1) \rightarrow (1, 1, 1): \quad 0, \quad \int_0^1 2 dy = 2, \quad \int_0^1 (2x + 4) dx = 5. \text{ Total} = 7 \checkmark$$

$$(c) \text{Parabolic path } y = x, z = x^2: \quad dy = dx, dz = 2x dx,$$

$$\nabla T \cdot d\mathbf{l} = (2x + 4x)dx + (4x + 2x^6)dx + 6x \cdot x^4 \cdot 2x dx = (10x + 14x^6)dx, \quad \int_0^1 (10x + 14x^6)dx = 5 + 2 = 7 \checkmark$$

Problem 1.33 Test the divergence theorem for $\mathbf{v} = xy\hat{\mathbf{x}} + 2yz\hat{\mathbf{y}} + 3zx\hat{\mathbf{z}}$ on the cube of side 2 (first octant).

$$\nabla \cdot \mathbf{v} = y + 2z + 3x, \quad \int (y + 2z + 3x) d\tau = \int_0^2 \int_0^2 (2y + 4z + 6) dy dz = \int_0^2 (8z + 16) dz = 48$$

Faces: (i) $x = 2$: $\iint 2y dy dz = 8$; (ii) $x = 0$: 0; (iii) $y = 2$: $\iint 4z dx dz = 16$; (iv) $y = 0$: 0; (v) $z = 2$: $\iint 6x dx dy = 24$; (vi) $z = 0$: 0.

$$\oint \mathbf{v} \cdot d\mathbf{a} = 8 + 16 + 24 = 48 \quad \checkmark$$

Problem 1.34 Test Stokes' theorem for the same $\mathbf{v}$ on the triangular region in the yz -plane with vertices $(0, 0, 0)$, $(0, 2, 0)$, $(0, 0, 2)$.

$\nabla \times \mathbf{v} = -2y\hat{\mathbf{x}} - 3z\hat{\mathbf{y}} - x\hat{\mathbf{z}}$, $d\mathbf{a} = dy dz \hat{\mathbf{x}}$:

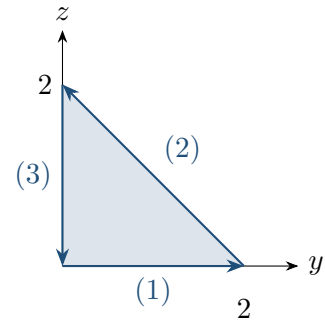
$$\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = \int_0^2 \int_0^{2-z} (-2y) dy dz = - \int_0^2 (2-z)^2 dz = -\frac{8}{3}$$

Line integral, $\mathbf{v} \cdot d\mathbf{l} = xy dx + 2yz dy + 3zx dz$: (1) $x = z = 0$: 0. (2) $x = 0$, $z = 2 - y$, $y: 2 \rightarrow 0$:

$$\int_2^0 2y(2-y) dy = - \left(2y^2 - \frac{2}{3}y^3 \right) \Big|_0^2 = -\frac{8}{3}$$

(3) $x = y = 0$: 0.

$$\oint \mathbf{v} \cdot d\mathbf{l} = -\frac{8}{3} \quad \checkmark$$



Problem 1.44 Delta-function integrals:

(a) $\int_2^6 (3x^2 - 2x - 1) \delta(x - 3) dx = 27 - 6 - 1 = 20$

(b) $\int_0^5 \cos x \delta(x - \pi) dx = \cos \pi = -1$

(c) $\int_0^3 x^3 \delta(x + 1) dx = 0$ ($x = -1$ lies outside the range)

(d) $\int_{-\infty}^{\infty} \ln(x + 3) \delta(x + 2) dx = \ln 1 = 0$

Problem 1.49 Evaluate $J = \int_V e^{-r} \left(\nabla \cdot \frac{\hat{\mathbf{r}}}{r^2} \right) d\tau$ over a sphere of radius R by two methods.

Method 1: $J = \int e^{-r} 4\pi \delta^3(\mathbf{r}) d\tau = 4\pi e^0 = 4\pi$.

Method 2 (integration by parts): with $\nabla(e^{-r}) = -e^{-r}\hat{\mathbf{r}}$,

$$\begin{aligned} J &= - \int_V \frac{\hat{\mathbf{r}}}{r^2} \cdot \nabla(e^{-r}) d\tau + \oint_S e^{-r} \frac{\hat{\mathbf{r}}}{r^2} \cdot d\mathbf{a} = 4\pi \int_0^R e^{-r} dr + e^{-R} \int \sin \theta d\theta d\phi \\ &= 4\pi(1 - e^{-R}) + 4\pi e^{-R} = 4\pi \end{aligned}$$

Problem 1.50

(a) $\mathbf{F}_1 = x^2 \hat{\mathbf{z}}$, $\mathbf{F}_2 = x \hat{\mathbf{x}} + y \hat{\mathbf{y}} + z \hat{\mathbf{z}}$.

$$\nabla \cdot \mathbf{F}_1 = 0, \quad \nabla \times \mathbf{F}_1 = -2x \hat{\mathbf{y}}; \quad \nabla \cdot \mathbf{F}_2 = 3, \quad \nabla \times \mathbf{F}_2 = \mathbf{0}$$

$\mathbf{F}_2$ is curl-free, so it is a gradient: $U_2 = \frac{1}{2}(x^2 + y^2 + z^2)$ gives $\mathbf{F}_2 = \nabla U_2$.

$\mathbf{F}_1$ is divergence-free, so it is a curl. We need $\partial_y A_z - \partial_z A_y = \partial_z A_x - \partial_x A_z = 0$ and $\partial_x A_y - \partial_y A_x = x^2$. Choose $A_y = x^3/3$, $A_x = A_z = 0$:

$$\mathbf{A}_1 = \frac{1}{3}x^3 \hat{\mathbf{y}} \quad (\mathbf{F}_1 = \nabla \times \mathbf{A}_1)$$

(b) $\mathbf{F}_3 = yz \hat{\mathbf{x}} + zx \hat{\mathbf{y}} + xy \hat{\mathbf{z}}$: $\nabla \cdot \mathbf{F}_3 = 0$ and $\nabla \times \mathbf{F}_3 = \hat{\mathbf{x}}(x - x) + \hat{\mathbf{y}}(y - y) + \hat{\mathbf{z}}(z - z) = \mathbf{0}$, so it is both a gradient and a curl. Scalar potential: $U_3 = xyz$. Vector potential (solving the three curl equations):

$$\mathbf{A}_3 = \frac{1}{4} \left[x(z^2 - y^2) \hat{\mathbf{x}} + y(x^2 - z^2) \hat{\mathbf{y}} + z(y^2 - x^2) \hat{\mathbf{z}} \right]$$

Chapter 2: Electrostatics

Theory

Q1. What is Coulomb's law? Write its expression.

The force on a test charge Q due to a single point charge q at rest a distance $\mathbf{r}$ away is proportional to the product of the charges and inversely proportional to the square of the separation:

$$\mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} \hat{\mathbf{r}}$$

Q2. What is an electric field? Write its expression.

The electric field at a point is the force per unit positive test charge placed at that point, $\mathbf{E} = \mathbf{F}/Q$. For a collection of point charges:

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i^2} \hat{\mathbf{r}}_i$$

Q3. What is Gauss's law? Write its expression.

The total electric flux through any closed surface equals the total charge enclosed divided by ϵ_0 .

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\epsilon_0} \quad (\text{integral form}), \quad \nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \quad (\text{differential form})$$

Q4. What is electric potential? Write its expression.

The electric potential at a point is the work done per unit positive test charge in bringing the charge from the reference point (usually infinity) to that point.

$$V(\mathbf{r}) = - \int_{\mathcal{O}}^{\mathbf{r}} \mathbf{E} \cdot d\mathbf{l}, \quad \mathbf{E} = -\nabla V$$

Q5. What are Poisson's and Laplace's equations?

They relate the potential to the charge distribution.

$$\text{Poisson: } \nabla^2 V = -\frac{\rho}{\epsilon_0} \quad (\text{where charge is present}), \quad \text{Laplace: } \nabla^2 V = 0 \quad (\text{charge-free region})$$

Q6. Explain boundary conditions in electrostatics.

They describe how the field and potential behave across a boundary carrying surface charge σ :

$$V_{\text{above}} = V_{\text{below}}, \quad E_{\text{above}}^{\parallel} = E_{\text{below}}^{\parallel}, \quad E_{\text{above}}^{\perp} - E_{\text{below}}^{\perp} = \frac{\sigma}{\epsilon_0}$$

Q7. Work required to move a charge.

To move a charge Q from $\mathbf{a}$ to $\mathbf{b}$:

$$W = \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{F} \cdot d\mathbf{l} = -Q \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l} = Q[V(\mathbf{b}) - V(\mathbf{a})]$$

Bringing it in from infinity (reference $V(\infty) = 0$): $W = QV(\mathbf{r})$.

Q8. Energy of a point-charge distribution.

The first charge q_1 takes no work (no field yet). The second takes $W_2 = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}}$. For many charges:

$$W = \frac{1}{4\pi\epsilon_0} \sum_{i < j} \frac{q_i q_j}{r_{ij}} = \frac{1}{2} \sum_i q_i V(\mathbf{r}_i)$$

Q9. Energy of a continuous charge distribution.

$$W = \frac{1}{2} \int \rho V d\tau = \frac{\epsilon_0}{2} \int_{\text{all space}} E^2 d\tau$$

Q10. Properties of conductors.

- (i) $\mathbf{E} = 0$ inside a conductor.
- (ii) $\rho = 0$ inside a conductor.
- (iii) Any net charge resides on the surface.
- (iv) A conductor is an equipotential.
- (v) Just outside a conductor, $\mathbf{E}$ is perpendicular to the surface.

Q11. Properties of capacitors.

A capacitor stores charge and electrostatic energy; it consists of two conductors separated by an insulator.

$$C = \frac{Q}{V}, \quad W = \frac{1}{2} CV^2 = \frac{1}{2} QV = \frac{Q^2}{2C}, \quad C_{\text{parallel plate}} = \frac{\epsilon_0 A}{d}$$

$$\text{Parallel: } C_{\text{eq}} = C_1 + C_2 + \dots, \quad \text{Series: } \frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$$

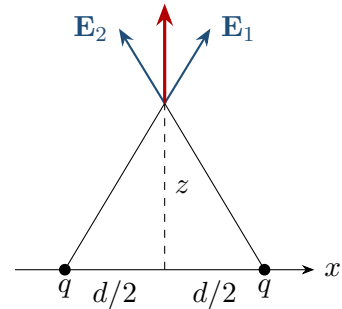
Examples

Example 2.1 Field at height z above the midpoint between two equal charges q separated by d .

The horizontal components cancel; the vertical components add:

$$E_z = 2 \cdot \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \cos \theta, \quad r = \sqrt{z^2 + (d/2)^2}, \quad \cos \theta = \frac{z}{r}$$

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{2qz}{[z^2 + (d/2)^2]^{3/2}} \hat{\mathbf{z}}$$



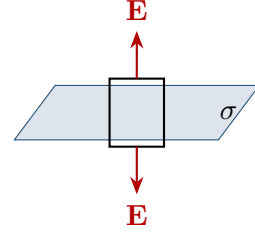
Example 2.3 Field outside a uniformly charged solid sphere (radius R , charge q). Take a Gaussian sphere of radius $r > R$. By symmetry $\mathbf{E}$ is radial with constant magnitude on it:

$$\oint \mathbf{E} \cdot d\mathbf{a} = |\mathbf{E}| 4\pi r^2 = \frac{q}{\epsilon_0} \Rightarrow \mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$$

Example 2.5 Field of an infinite plane with uniform surface charge σ .

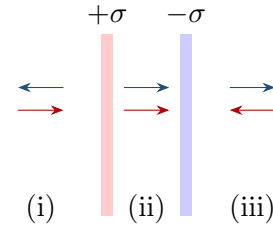
Use a Gaussian pillbox extending equally above and below the plane, lid area A . $Q_{\text{enc}} = \sigma A$ and by symmetry only the two lids contribute:

$$2A|\mathbf{E}| = \frac{\sigma A}{\epsilon_0} \Rightarrow \mathbf{E} = \frac{\sigma}{2\epsilon_0} \hat{\mathbf{n}}$$



Example 2.6 Two infinite parallel planes with charge densities $+\sigma$ (left) and $-\sigma$ (right). Find the field (i) left of both, (ii) between, (iii) right of both.

The left plate produces $\sigma/2\epsilon_0$ pointing away from it: leftward in (i), rightward in (ii) and (iii). The right plate produces $\sigma/2\epsilon_0$ pointing toward it: rightward in (i) and (ii), leftward in (iii).



$$E_{\text{(ii)}} = \frac{\sigma}{\epsilon_0} \text{ (to the right),} \quad E_{\text{(i)}} = E_{\text{(iii)}} = 0$$

Example 2.7 Potential inside and outside a spherical shell of radius R with total charge q (reference at infinity).

The field outside is $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$; inside it is zero. For $r \geq R$:

$$V(r) = - \int_{\infty}^r \frac{1}{4\pi\epsilon_0} \frac{q}{r'^2} dr' = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

For $r \leq R$ the field is zero inside, so

$$V(r) = - \frac{1}{4\pi\epsilon_0} \int_{\infty}^R \frac{q}{r'^2} dr' - \int_R^r 0 dr' = \frac{1}{4\pi\epsilon_0} \frac{q}{R} \quad (\text{constant})$$

Example 2.9 Energy of a uniformly charged spherical shell (charge q , radius R).

Solution 1: On the surface $V = \frac{1}{4\pi\epsilon_0} \frac{q}{R}$, so

$$W = \frac{1}{2} \int \sigma V da = \frac{1}{2} \frac{1}{4\pi\epsilon_0} \frac{q}{R} \int \sigma da = \frac{1}{8\pi\epsilon_0} \frac{q^2}{R}$$

Solution 2: $E = 0$ inside and $E^2 = \frac{q^2}{(4\pi\epsilon_0)^2 r^4}$ outside:

$$W = \frac{\epsilon_0}{2(4\pi\epsilon_0)^2} \int_{\text{outside}} \frac{q^2}{r^4} r^2 \sin \theta dr d\theta d\phi = \frac{q^2 4\pi}{32\pi^2 \epsilon_0} \int_R^{\infty} \frac{dr}{r^2} = \frac{1}{8\pi\epsilon_0} \frac{q^2}{R}$$

Example 2.11 Capacitance of a parallel-plate capacitor (plate area A , separation d). Put $+Q$ on the top plate and $-Q$ on the bottom; the charge spreads uniformly with $\sigma = Q/A$. The field between is $E = \sigma/\epsilon_0 = Q/(A\epsilon_0)$, so $V = Ed = \frac{Qd}{A\epsilon_0}$ and

$$C = \frac{Q}{V} = \frac{A\epsilon_0}{d}$$

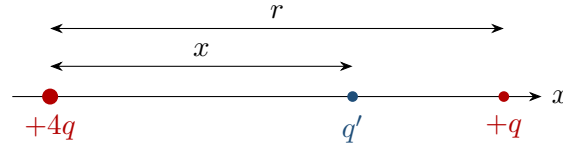
Example 2.12 Capacitance of two concentric spherical metal shells of radii $a < b$. Put $+Q$ on the inner and $-Q$ on the outer shell. Between them $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{\mathbf{r}}$, so

$$V = - \int_b^a \mathbf{E} \cdot d\mathbf{l} = - \frac{Q}{4\pi\epsilon_0} \int_b^a \frac{dr}{r^2} = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right)$$

$$C = \frac{Q}{V} = 4\pi\epsilon_0 \frac{ab}{b-a}$$

Problems

Problem 24 Two free charges $+4q$ and $+q$ are a distance r apart. Find the force between them, and the magnitude, sign and location of a third charge q' such that (i) there is no force on q' ; (ii) there is no force on any of the three charges (equilibrium).



Force between $+4q$ and $+q$ (repulsive):

$$\mathbf{F}_{4q} = \frac{1}{4\pi\epsilon_0} \frac{4q^2}{r^2} (-\hat{\mathbf{x}}), \quad \mathbf{F}_q = \frac{1}{4\pi\epsilon_0} \frac{4q^2}{r^2} (+\hat{\mathbf{x}})$$

(i) No force on q' . Place q' at distance x from $+4q$, between the charges. Balancing:

$$\frac{1}{4\pi\epsilon_0} \frac{4qq'}{x^2} = \frac{1}{4\pi\epsilon_0} \frac{qq'}{(r-x)^2} \Rightarrow 4(r-x)^2 = x^2 \Rightarrow 3x^2 - 8rx + 4r^2 = 0$$

$$x = \frac{8r \pm \sqrt{64r^2 - 48r^2}}{6} = 2r \text{ or } \frac{2r}{3}$$

$x = 2r$ lies outside the charges where both forces point the same way, so it is rejected: $x = \frac{2r}{3}$. The result holds for either sign of q' (if q' is negative both forces are attractive and still balance).

(ii) Whole system in equilibrium. If q' is positive, the outer charges can't be in equilibrium, so q' must be negative. For no force on $+4q$:

$$\frac{1}{4\pi\epsilon_0} \frac{4q|q'|}{x^2} = \frac{1}{4\pi\epsilon_0} \frac{4q \cdot q}{r^2} \Rightarrow |q'| = q \frac{x^2}{r^2} = q \cdot \frac{4}{9}$$

$$q' = -\frac{4q}{9} \text{ placed at } x = \frac{2r}{3} \text{ from } +4q$$

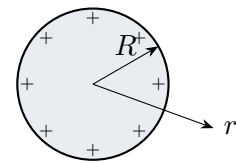
Problem 61 Potential inside and outside a conducting sphere of radius R carrying charge q .

Outside, $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$. The charge does not extend to infinity, so take infinity as reference:

$$V_{\text{out}} = - \int_{\infty}^r \frac{q}{4\pi\epsilon_0 r'^2} dr' = \frac{q}{4\pi\epsilon_0 r}$$

Inside, the field is zero, so the work from infinity to the surface is all there is:

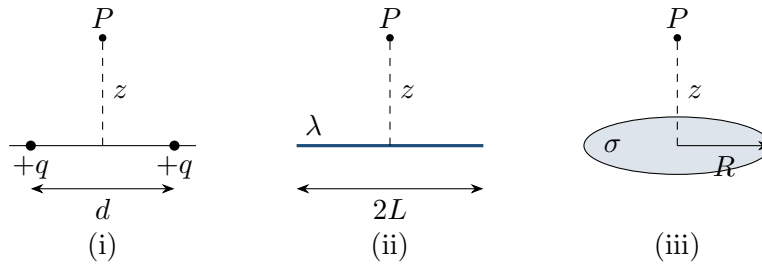
$$V_{\text{in}} = - \int_{\infty}^R \frac{q}{4\pi\epsilon_0 r'^2} dr' - \int_R^r 0 dr' = \frac{q}{4\pi\epsilon_0 R}$$



Problem 63 Potential at distance s from an infinite straight wire with uniform line charge λ . The field is $\mathbf{E} = \frac{\lambda}{2\pi\epsilon_0 s} \hat{\mathbf{s}}$. The charge extends to infinity, so we can't use infinity as reference; use a finite point at distance a :

$$V = - \int_a^s \frac{\lambda}{2\pi\epsilon_0 s'} ds' = - \frac{\lambda}{2\pi\epsilon_0} \ln \frac{s}{a} \Rightarrow V = \frac{\lambda}{2\pi\epsilon_0} \ln \left(\frac{a}{s} \right)$$

Problem 65 Potential (and field) at height z above the centre of three configurations:



(i) Two charges $+q$ separated by d . Each is at distance $\sqrt{z^2 + d^2/4}$:

$$V = \frac{2q}{4\pi\epsilon_0\sqrt{z^2 + d^2/4}} = \frac{q}{2\pi\epsilon_0\sqrt{z^2 + d^2/4}}, \quad \mathbf{E} = -\frac{\partial V}{\partial z}\hat{\mathbf{z}} = \frac{qz}{2\pi\epsilon_0(z^2 + d^2/4)^{3/2}}\hat{\mathbf{z}}$$

(ii) Line segment of length $2L$ with charge λ . With $\mathbf{r} = \sqrt{z^2 + x^2}$:

$$V = \frac{2}{4\pi\epsilon_0} \int_0^L \frac{\lambda dx}{\sqrt{z^2 + x^2}} = \frac{\lambda}{2\pi\epsilon_0} \ln \left[x + \sqrt{z^2 + x^2} \right]_0^L = \frac{\lambda}{2\pi\epsilon_0} \ln \left[\frac{L + \sqrt{L^2 + z^2}}{z} \right]$$

(using $\int \frac{dx}{\sqrt{a^2 + x^2}} = \ln(x + \sqrt{a^2 + x^2})$). Then

$$\mathbf{E} = -\frac{\partial V}{\partial z}\hat{\mathbf{z}} = \frac{\lambda}{2\pi\epsilon_0} \left[\frac{1}{z} - \frac{z}{\sqrt{z^2 + L^2}(L + \sqrt{z^2 + L^2})} \right] \hat{\mathbf{z}} = \frac{\lambda L}{2\pi\epsilon_0 z\sqrt{z^2 + L^2}} \hat{\mathbf{z}}$$

(iii) Disk of radius R with surface charge σ . A ring of radius r' has $dq = 2\pi r' dr' \sigma$:

$$V = \frac{\sigma}{4\pi\epsilon_0} \int_0^R \frac{2\pi r' dr'}{\sqrt{z^2 + r'^2}} \quad (t = z^2 + r'^2, \quad dt = 2r' dr')$$

$$V = \frac{\sigma}{2\epsilon_0} \left[\sqrt{z^2 + R^2} - z \right]$$

At the centre ($z = 0$): $V_{\text{centre}} = \frac{\sigma R}{2\epsilon_0}$. At the rim of the disk (a separate calculation): $V_{\text{edge}} = \frac{\sigma R}{\pi\epsilon_0}$.

Problem 66 Is $\mathbf{E} = (12x^2 - y^2)\hat{\mathbf{x}} - 2xy\hat{\mathbf{y}}$ conservative? If so find V .

$$\nabla \times \mathbf{E} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial_x & \partial_y & \partial_z \\ 12x^2 - y^2 & -2xy & 0 \end{vmatrix} = \hat{\mathbf{z}}[-2y - (-2y)] = \mathbf{0}$$

so $\mathbf{E}$ is conservative. Since $\mathbf{E} = \nabla(4x^3 - xy^2)$,

$$V = -\int \mathbf{E} \cdot d\mathbf{l} = xy^2 - 4x^3 + C$$

Problem 98 A parallel-plate capacitor has plates at $x = 0$ (grounded) and $x = L$ (potential V_0), with uniform charge density ρ in between. Find V and $\mathbf{E}$ between the plates.

Poisson's equation in one dimension:

$$\frac{d^2V}{dx^2} = -\frac{\rho}{\epsilon_0} \Rightarrow V = -\frac{\rho x^2}{2\epsilon_0} + Ax + B$$

$$V(0) = 0 \Rightarrow B = 0; \quad V(L) = V_0 \Rightarrow A = \frac{V_0}{L} + \frac{\rho L}{2\epsilon_0}.$$

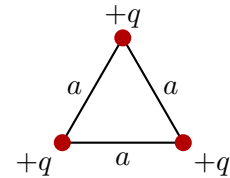
$$V(x) = -\frac{\rho x^2}{2\epsilon_0} + \frac{V_0 x}{L} + \frac{\rho L x}{2\epsilon_0}$$

$$\mathbf{E} = -\frac{dV}{dx} \hat{\mathbf{x}} = \left[\frac{\rho x}{\epsilon_0} - \frac{V_0}{L} - \frac{\rho L}{2\epsilon_0} \right] \hat{\mathbf{x}}$$

Problem 81 Energy of three charges $+q$ at the corners of an equilateral triangle of side a .

$$W = \frac{1}{8\pi\epsilon_0} \sum_{i \neq j} \frac{q_i q_j}{r_{ij}} = \frac{1}{8\pi\epsilon_0} \cdot \frac{6q^2}{a}$$

Equivalently, build it up: $W_1 = 0$, $W_2 = qV_1 = \frac{q^2}{4\pi\epsilon_0 a}$, $W_3 = q(V_1 + V_2) = \frac{2q^2}{4\pi\epsilon_0 a}$.



$$W = \frac{3q^2}{4\pi\epsilon_0 a}$$

Chapter 3: Potentials

Theory

Q1. Explain the method of images for finding the electric potential.

A charge $+q$ near a grounded conducting plane induces a charge distribution on the conductor. Instead of calculating the induced charge directly, we replace its effect with an imaginary *image charge* $-q$ placed symmetrically behind the plane. The image charge is not real; it is a mathematical device chosen so that the boundary condition ($V = 0$ on the grounded conductor) is satisfied. By the uniqueness theorem, the resulting potential is then the correct one in the region above the plane.

Q2. Using the multipole expansion, find the approximate potential far from a dipole. (also Example 3.10)

Let $\mathbf{r}_{\pm}$ be the distances from $\pm q$:

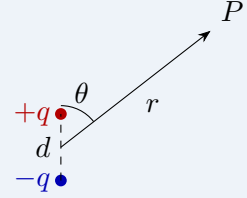
$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{\mathbf{r}_+} - \frac{q}{\mathbf{r}_-} \right),$$

$$\mathbf{r}_{\pm}^2 = r^2 + \left(\frac{d}{2}\right)^2 \mp rd \cos \theta = r^2 \left(1 \mp \frac{d}{r} \cos \theta + \frac{d^2}{4r^2} \right).$$

For $r \gg d$:

$$\frac{1}{\mathbf{r}_{\pm}} \cong \frac{1}{r} \left(1 \mp \frac{d}{r} \cos \theta \right)^{-1/2} \cong \frac{1}{r} \left(1 \pm \frac{d}{2r} \cos \theta \right) \Rightarrow \frac{1}{\mathbf{r}_+} - \frac{1}{\mathbf{r}_-} \cong \frac{d}{r^2} \cos \theta$$

$$V(\mathbf{r}) \cong \frac{1}{4\pi\epsilon_0} \frac{qd \cos \theta}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \hat{\mathbf{r}}}{r^2}$$



Q3. Using the multipole expansion, expand the potential of any localized charge distribution in powers of $1/r$.

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{1}{\mathbf{r}} \rho(\mathbf{r}') d\tau'$$

By the law of cosines,

$$\mathbf{r}^2 = r^2 + r'^2 - 2rr' \cos \alpha = r^2 \left[1 + \left(\frac{r'}{r}\right)^2 - 2\left(\frac{r'}{r}\right) \cos \alpha \right],$$

so $\mathbf{r} = r\sqrt{1 + \epsilon}$ with $\epsilon = \left(\frac{r'}{r}\right) \left(\frac{r'}{r} - 2 \cos \alpha\right)$.

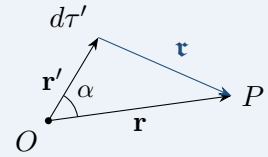
For points well outside the distribution $\epsilon \ll 1$ and

$$\frac{1}{\mathbf{r}} = \frac{1}{r} (1 + \epsilon)^{-1/2} = \frac{1}{r} \left(1 - \frac{1}{2}\epsilon + \frac{3}{8}\epsilon^2 - \frac{5}{16}\epsilon^3 + \dots \right)$$

Collecting powers of r'/r :

$$\frac{1}{\mathbf{r}} = \frac{1}{r} \left[1 + \left(\frac{r'}{r}\right) \cos \alpha + \left(\frac{r'}{r}\right)^2 \left(\frac{3 \cos^2 \alpha - 1}{2} \right) + \left(\frac{r'}{r}\right)^3 \left(\frac{5 \cos^3 \alpha - 3 \cos \alpha}{2} \right) + \dots \right]$$

The coefficients are the Legendre polynomials, $\frac{1}{\mathbf{r}} = \frac{1}{r} \sum_{n=0}^{\infty} \left(\frac{r'}{r}\right)^n P_n(\cos \alpha)$, hence



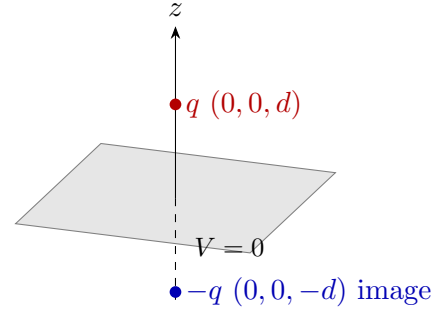
$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{n=0}^{\infty} \frac{1}{r^{n+1}} \int (r')^n P_n(\cos \alpha) \rho(\mathbf{r}') d\tau'$$

The $n = 0$ term is the monopole ($\sim 1/r$), $n = 1$ the dipole ($\sim 1/r^2$), $n = 2$ the quadrupole ($\sim 1/r^3$), and so on.

Problems

Problem 86 A charge q is at height d above an infinite grounded conducting plane. Find (i) the potential and (ii) the field in the region above the plane, the force on the charge, and the energy.

Boundary conditions: $V = 0$ at $z = 0$, and $V \rightarrow 0$ far away ($x^2 + y^2 + z^2 \gg d^2$). Replace the problem with q at $(0, 0, d)$ and an image $-q$ at $(0, 0, -d)$. For $z \geq 0$ the potential is that of the two point charges (below).



$$V(x, y, z) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{\sqrt{x^2 + y^2 + (z-d)^2}} - \frac{q}{\sqrt{x^2 + y^2 + (z+d)^2}} \right]$$

The field ($\mathbf{E} = -\nabla V$) for $z > 0$:

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \left[\frac{q(x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + (z-d)\hat{\mathbf{z}})}{[x^2 + y^2 + (z-d)^2]^{3/2}} - \frac{q(x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + (z+d)\hat{\mathbf{z}})}{[x^2 + y^2 + (z+d)^2]^{3/2}} \right]$$

Its z -component is

$$E_z = \frac{q}{4\pi\epsilon_0} \left[\frac{z-d}{[x^2 + y^2 + (z-d)^2]^{3/2}} - \frac{z+d}{[x^2 + y^2 + (z+d)^2]^{3/2}} \right],$$

and at the plane ($z = 0$) only this normal component survives. Below the plane, $\mathbf{E} = 0$.

Induced surface charge: $\sigma = -\epsilon_0 \frac{\partial V}{\partial z} \Big|_{z=0} = \frac{-qd}{2\pi(x^2 + y^2 + d^2)^{3/2}}$ (total $-q$).

Force on q (attraction toward the plane, same as from the image):

$$\mathbf{F} = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{(2d)^2} \hat{\mathbf{z}}$$

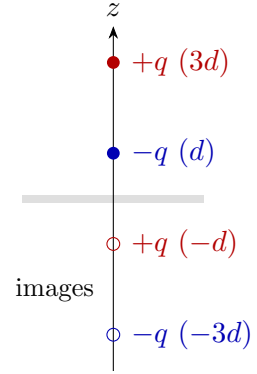
Energy (work to bring q in from infinity; only half the energy of a real two-charge system because only the real charge moves):

$$W = -\int_{\infty}^d \mathbf{F} \cdot d\mathbf{l} = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{4d}$$

Problem 91 The xy -plane is a grounded conductor, with $-q$ at $z = d$ and $+q$ at $z = 3d$. Find the force on $+q$ and the energy of the configuration.

Image setup: $+q$ at $-d$ and $-q$ at $-3d$. Forces on $+q$ at $3d$:

- from $-q$ at d (distance $2d$, attractive): $\frac{1}{4\pi\epsilon_0} \frac{q^2}{4d^2} (-\hat{\mathbf{z}})$
- from $+q$ at $-d$ (distance $4d$, repulsive): $\frac{1}{4\pi\epsilon_0} \frac{q^2}{16d^2} (+\hat{\mathbf{z}})$
- from $-q$ at $-3d$ (distance $6d$, attractive): $\frac{1}{4\pi\epsilon_0} \frac{q^2}{36d^2} (-\hat{\mathbf{z}})$



$$\mathbf{F} = -\frac{q^2}{4\pi\epsilon_0 d^2} \left[\frac{1}{4} - \frac{1}{16} + \frac{1}{36} \right] \hat{\mathbf{z}} = -\frac{q^2}{4\pi\epsilon_0 d^2} \cdot \frac{36 - 9 + 4}{144} \hat{\mathbf{z}}$$

$$\mathbf{F} = -\frac{1}{4\pi\epsilon_0} \frac{31 q^2}{144 d^2} \hat{\mathbf{z}}$$

Energy. Use $W = \frac{1}{2} \sum_i q_i V_i$ over the *real* charges, where V_i is the potential at q_i due to all other charges (real and image):

$$V(3d) = \frac{q}{4\pi\epsilon_0 d} \left[-\frac{1}{2} + \frac{1}{4} - \frac{1}{6} \right] = -\frac{5}{12} \frac{q}{4\pi\epsilon_0 d}, \quad V(d) = \frac{q}{4\pi\epsilon_0 d} \left[\frac{1}{2} + \frac{1}{2} - \frac{1}{4} \right] = \frac{3}{4} \frac{q}{4\pi\epsilon_0 d}$$

$$W = \frac{1}{2} \left[q \left(-\frac{5}{12} \right) + (-q) \left(\frac{3}{4} \right) \right] \frac{q}{4\pi\epsilon_0 d}$$

$$W = -\frac{7 q^2}{12 (4\pi\epsilon_0) d}$$

Chapter 4: Electric Fields in Matter

Theory

Q1. Differences between conductors, insulators and dielectrics.

Property	Conductor	Insulator	Dielectric
Free charges	Large number	Very few / none	Very few / none
Charge motion	Move freely	Cannot move freely	Bound charges shift slightly
Field inside	$E = 0$ (static)	Can be non-zero	Can be non-zero
Polarization	Not normally described by bound charge	May be very small	Significant
In external field	Free charges redistribute	Little charge movement	Molecules/atoms become polarized
Examples	Copper, aluminium	Glass, rubber, plastic	Glass, mica, ceramic

Q2. Explain electric polarization.

When a dielectric is placed in an external field, the positive and negative charges in its molecules are pushed in opposite directions, so their centres separate slightly. This creates tiny electric dipoles throughout the material — electric polarization. $\mathbf{P}$ is the dipole moment per unit volume; for a linear dielectric

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}.$$

Q3. Explain bound charges using the potential of a polarized object.

Bound charges are tied to specific locations in the material; they do not flow as current, but they do affect the overall field. The potential of a polarized object is the sum of dipole potentials:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\mathbf{P}(\mathbf{r}') \cdot \hat{\mathbf{r}}}{r'^2} d\tau'$$

Using $\nabla' \left(\frac{1}{r} \right) = \frac{\hat{\mathbf{r}}}{r^2}$ and integrating by parts:

$$V = \frac{1}{4\pi\epsilon_0} \int_V \mathbf{P} \cdot \nabla' \left(\frac{1}{r} \right) d\tau' = \frac{1}{4\pi\epsilon_0} \left[\int_V \nabla' \cdot \left(\frac{\mathbf{P}}{r} \right) d\tau' - \int_V \frac{1}{r} (\nabla' \cdot \mathbf{P}) d\tau' \right]$$

$$V = \frac{1}{4\pi\epsilon_0} \oint_S \frac{1}{r} \mathbf{P} \cdot d\mathbf{a}' - \frac{1}{4\pi\epsilon_0} \int_V \frac{1}{r} (\nabla' \cdot \mathbf{P}) d\tau'$$

The first term looks like the potential of a surface charge and the second like a volume charge:

$$\sigma_b \equiv \mathbf{P} \cdot \hat{\mathbf{n}}, \quad \rho_b \equiv -\nabla \cdot \mathbf{P}$$

Q4. Explain Gauss's law in the presence of dielectrics.

Inside a dielectric the total charge density is $\rho = \rho_b + \rho_f$. Gauss's law gives

$$\epsilon_0 \nabla \cdot \mathbf{E} = \rho = \rho_b + \rho_f = -\nabla \cdot \mathbf{P} + \rho_f \Rightarrow \nabla \cdot (\epsilon_0 \mathbf{E} + \mathbf{P}) = \rho_f,$$

where $\mathbf{E}$ is now the *total* field. Defining the electric displacement $\mathbf{D} \equiv \epsilon_0 \mathbf{E} + \mathbf{P}$:

$$\nabla \cdot \mathbf{D} = \rho_f, \quad \oint \mathbf{D} \cdot d\mathbf{a} = Q_{f,\text{enc}}$$

Q5. Explain electric susceptibility, permittivity and dielectric constant.

For a linear dielectric $\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}$, where the *electric susceptibility* χ_e measures how easily the material polarizes. Then

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \epsilon_0 \chi_e \mathbf{E} = \epsilon_0 (1 + \chi_e) \mathbf{E} = \epsilon \mathbf{E}, \quad \epsilon = \epsilon_0 (1 + \chi_e)$$

ϵ is the *permittivity* of the material (for vacuum $\epsilon = \epsilon_0$). The *dielectric constant* is the relative permittivity:

$$\epsilon_r = \frac{\epsilon}{\epsilon_0} = 1 + \chi_e$$

Problems

Problem 109 A primitive atom model: point nucleus $+q$ surrounded by a uniformly charged spherical cloud $-q$ of radius a . Find the atomic polarizability.

In an external field $\mathbf{E}$ the nucleus shifts by d from the cloud's centre. Equilibrium occurs where the field of the cloud at distance d balances $\mathbf{E}$. Inside a uniform sphere, with $\rho = \frac{q}{\frac{4}{3}\pi a^3}$:

$$E_e = \frac{\rho d}{3\epsilon_0} = \frac{1}{4\pi\epsilon_0} \frac{qd}{a^3}$$

So $p = qd = (4\pi\epsilon_0 a^3)E$. Comparing with $\mathbf{p} = \alpha \mathbf{E}$:

$$\alpha = 4\pi\epsilon_0 a^3 = 3\epsilon_0 v$$

where v is the volume of the atom.

Problem 112 A sphere of radius R carries polarization $\mathbf{P}(\mathbf{r}) = k\mathbf{r}$ (k constant). (a) Find σ_b and ρ_b . (b) Find the field inside and outside.

(a) At the surface $\hat{\mathbf{n}} = \hat{\mathbf{r}}$:

$$\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}} = kR, \quad \rho_b = -\nabla \cdot \mathbf{P} = -\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \cdot kr) = -3k$$

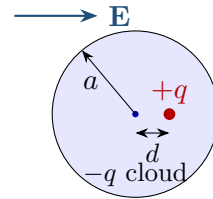
(b) *Inside* ($r < R$), only the volume charge is enclosed:

$$Q_{\text{enc}} = -3k \cdot \frac{4}{3}\pi r^3 = -4\pi k r^3, \quad E \cdot 4\pi r^2 = \frac{-4\pi k r^3}{\epsilon_0}$$

$$\mathbf{E}_{\text{in}} = -\frac{k}{\epsilon_0} \mathbf{r}$$

Outside: total bound charge = $\sigma_b(4\pi R^2) + \rho_b(\frac{4}{3}\pi R^3) = 4\pi k R^3 - 4\pi k R^3 = 0$, so

$$\mathbf{E}_{\text{out}} = 0$$



Problem 114 A long straight wire with line charge λ is surrounded by rubber insulation (dielectric constant ϵ_r) out to radius a . Find $\mathbf{D}$ and $\mathbf{E}$ inside and outside the rubber.

Gaussian cylinder of radius s , length l : $Q_{f,enc} = \lambda l$, so for all s

$$D(2\pi sl) = \lambda l \Rightarrow \mathbf{D} = \frac{\lambda}{2\pi s} \hat{\mathbf{s}}$$

Inside the rubber ($s < a$), $\mathbf{D} = \epsilon \mathbf{E}$:

$$\mathbf{E}_{in} = \frac{\mathbf{D}}{\epsilon_0 \epsilon_r} = \frac{\lambda}{2\pi \epsilon_0 \epsilon_r s} \hat{\mathbf{s}}$$

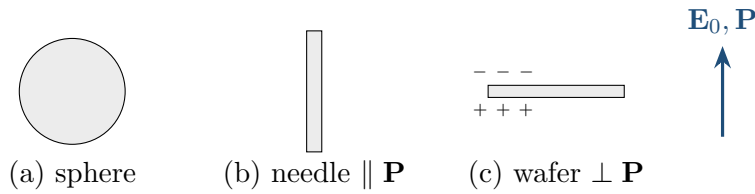


Outside ($s > a$), $\epsilon_r = 1$:

$$\mathbf{E}_{out} = \frac{\lambda}{2\pi \epsilon_0 s} \hat{\mathbf{s}}$$

Problem 118 Inside a large dielectric the field is $\mathbf{E}_0$ and displacement $\mathbf{D}_0 = \epsilon_0 \mathbf{E}_0 + \mathbf{P}$. Find $\mathbf{E}$ and $\mathbf{D}$ inside a small cavity that is (a) spherical; (b) a long needle parallel to $\mathbf{P}$; (c) a thin wafer perpendicular to $\mathbf{P}$. Assume $\mathbf{P}$ is unchanged by the cavity.

Idea (superposition): hollowing out a cavity is equivalent to adding a small object of the same shape with polarization $-\mathbf{P}$. The field in the cavity is $\mathbf{E}_0$ plus the field of that object.



(a) *Spherical cavity.* A uniformly polarized sphere with polarization $-\mathbf{P}$ produces a field $+\mathbf{P}/3\epsilon_0$ inside it:

$$\mathbf{E} = \mathbf{E}_0 + \frac{\mathbf{P}}{3\epsilon_0}, \quad \mathbf{D} = \epsilon_0 \mathbf{E} = \epsilon_0 \mathbf{E}_0 + \frac{\mathbf{P}}{3} = \mathbf{D}_0 - \frac{2}{3} \mathbf{P}$$

(b) *Needle-shaped cavity* (parallel to $\mathbf{P}$). The bound charge sits only on the tiny end faces, far apart, so its field is negligible:

$$\mathbf{E} = \mathbf{E}_0, \quad \mathbf{D} = \epsilon_0 \mathbf{E}_0 = \mathbf{D}_0 - \mathbf{P}$$

(c) *Wafer-shaped cavity* (perpendicular to $\mathbf{P}$). The faces act like a parallel-plate capacitor with $\sigma_b = \pm P$, adding a field $\mathbf{P}/\epsilon_0$:

$$\mathbf{E} = \mathbf{E}_0 + \frac{\mathbf{P}}{\epsilon_0}, \quad \mathbf{D} = \epsilon_0 \mathbf{E}_0 + \mathbf{P} = \mathbf{D}_0$$

Chapter 5: Magnetostatics

Theory

Q1. Magnetic force on (i) a moving charge and (ii) a current-carrying wire.

(i) A charge q moving with velocity $\mathbf{v}$ in a field $\mathbf{B}$ experiences

$$\mathbf{F} = q(\mathbf{v} \times \mathbf{B}).$$

(ii) On a small element $d\mathbf{l}$ of a wire carrying current I : $d\mathbf{F} = I(d\mathbf{l} \times \mathbf{B})$. For a straight wire of length $\mathbf{L}$ in a uniform field: $\mathbf{F} = I(\mathbf{L} \times \mathbf{B})$.

Q2. Explain the Biot–Savart law.

It gives the magnetic field produced by a steady current:

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{I} \times \hat{\mathbf{r}}}{r^2} dl' = \frac{\mu_0 I}{4\pi} \int \frac{d\mathbf{l}' \times \hat{\mathbf{r}}}{r^2}$$

It is the magnetostatic analogue of Coulomb's law.

Q3. Significance of the divergence of the magnetic field.

$$\nabla \cdot \mathbf{B} = 0$$

The magnetic field has zero divergence everywhere: field lines never begin or end, they form closed loops. Equivalently, no magnetic monopoles have been observed.

Q4. Significance of the curl of the magnetic field (Ampère's law).

The circulation of $\mathbf{B}$ around a closed path is proportional to the current enclosed:

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}, \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J}$$

Current is the source of the circulation (curl) of the magnetic field.

Q5. Explain the magnetic vector potential and compare it with the electrostatic scalar potential.

Because $\nabla \cdot \mathbf{B} = 0$, we can write $\mathbf{B} = \nabla \times \mathbf{A}$. Adding the gradient of any scalar to $\mathbf{A}$ does not change $\mathbf{B}$ (gauge freedom):

$$\mathbf{A}' = \mathbf{A} + \nabla \lambda$$

Choosing $\nabla \cdot \mathbf{A} = 0$ (Coulomb gauge), Ampère's law becomes

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} = \mu_0 \mathbf{J} \Rightarrow \nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}, \quad \mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}')}{r} d\tau'$$

This is three Poisson equations, one per component. In electrostatics, $\mathbf{E} = -\nabla V$ with V a scalar satisfying $\nabla^2 V = -\rho/\epsilon_0$; the field comes from the gradient of V , whereas $\mathbf{B}$ comes from the curl of the vector $\mathbf{A}$.

Q6. Boundary conditions in magnetostatics.

The magnetic field is discontinuous across a surface current $\mathbf{K}$:

$$B_{\text{above}}^{\perp} = B_{\text{below}}^{\perp}, \quad B_{\text{above}}^{\parallel} - B_{\text{below}}^{\parallel} = \mu_0 K$$

The vector potential is continuous across any boundary: $\mathbf{A}_{\text{above}} = \mathbf{A}_{\text{below}}$.

Q7. Multipole expansion of the vector potential.

For a current loop, valid at distant points,

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0 I}{4\pi} \oint \frac{1}{r} d\mathbf{l}' = \frac{\mu_0 I}{4\pi} \sum_{n=0}^{\infty} \frac{1}{r^{n+1}} \oint (r')^n P_n(\cos \alpha) d\mathbf{l}'$$

The monopole term ($\sim 1/r$) is always zero, the dipole term goes as $1/r^2$, the quadrupole as $1/r^3$. The dipole term is

$$\mathbf{A}_{\text{dip}}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{\mathbf{m} \times \hat{\mathbf{r}}}{r^2}, \quad \mathbf{m} = I \int d\mathbf{a} = I\mathbf{a}$$

Problems

Problem 1 An electron enters a uniform field $B = 0.3 \text{ T}$ at 45° to the field with speed $2 \times 10^6 \text{ m/s}$. Find the radius and pitch of its helical path.

$$v_{\perp} = v \sin 45^\circ = \sqrt{2} \times 10^6 \text{ m/s}$$

$$v_{\parallel} = v \cos 45^\circ = \sqrt{2} \times 10^6 \text{ m/s}$$

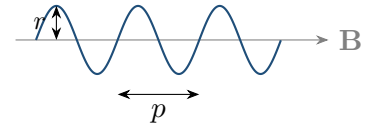
$v_{\perp}$ gives circular motion; $v_{\parallel}$ gives forward drift.

$$qv_{\perp}B = \frac{mv_{\perp}^2}{r} \Rightarrow r = \frac{mv_{\perp}}{qB} = \frac{9.1 \times 10^{-31} \times \sqrt{2} \times 10^6}{1.6 \times 10^{-19} \times 0.3}$$

$$r \approx 26.8 \text{ } \mu\text{m}$$

Period $\Delta t = 2\pi r/v_{\perp}$, so pitch $p = v_{\parallel} \Delta t = 2\pi r (v_{\parallel}/v_{\perp}) = 2\pi r$:

$$p = 2\pi \times 26.8 \text{ } \mu\text{m} \approx 168 \text{ } \mu\text{m}$$



Problem 2 (a) Current I is uniformly distributed over a wire of circular cross-section, radius a . Find $\mathbf{J}$. (b) If instead $J = ks$ (proportional to distance from the axis), find the total current.

(a) $J = \frac{I}{\pi a^2}$.

(b) The current density is non-uniform, so integrate over the cross-section:

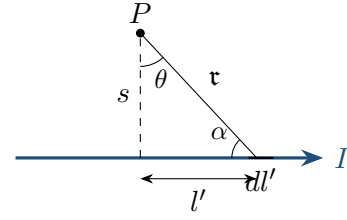
$$I = \int J da_{\perp} = \int_0^{2\pi} \int_0^a (ks) s ds d\phi = 2\pi k \frac{a^3}{3} \Rightarrow I = \frac{2\pi k a^3}{3}$$

Problem 3 Magnetic field at distance s from a straight wire carrying steady current I (Biot–Savart).

$$B = \frac{\mu_0 I}{4\pi} \int \frac{dl' \sin \alpha}{r^2}, \quad \sin \alpha = \cos \theta$$

With $l' = s \tan \theta$, $dl' = s \sec^2 \theta d\theta$ and $r = s \sec \theta$:

$$B = \frac{\mu_0 I}{4\pi} \int_{\theta_1}^{\theta_2} \frac{\cos^2 \theta}{s^2} \cos \theta \cdot s \sec^2 \theta d\theta = \frac{\mu_0 I}{4\pi s} \int_{\theta_1}^{\theta_2} \cos \theta d\theta$$

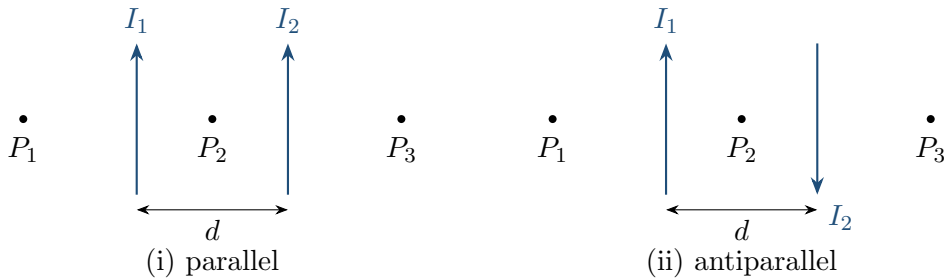


$$B = \frac{\mu_0 I}{4\pi s} (\sin \theta_2 - \sin \theta_1) \quad (\text{finite wire})$$

For an infinite wire $\theta_1 = -90^\circ$, $\theta_2 = +90^\circ$:

$$\mathbf{B} = \frac{\mu_0 I}{2\pi s} \hat{\phi}$$

Problem 4 Force per unit length between two long parallel wires a distance d apart, and the field at P_1 (distance d left of wire 1), P_2 (midway) and P_3 (distance d right of wire 2), for (i) parallel and (ii) antiparallel currents.



The field of wire 1 at wire 2 is $B_1 = \mu_0 I_1 / 2\pi d$. The force on wire 2 is $\mathbf{F}_2 = I_2 \int d\mathbf{l}_2 \times \mathbf{B}_1$, so per unit length

$$f = \frac{\mu_0 I_1 I_2}{2\pi d}$$

By Newton's third law the force on wire 1 has the same magnitude. The force is $\propto I_1 I_2$ and $\propto 1/d$.

- (i) Parallel currents *attract* (wire 2 is pushed left, wire 1 right).
- (ii) Antiparallel currents *repel*; same magnitude, directions reversed.

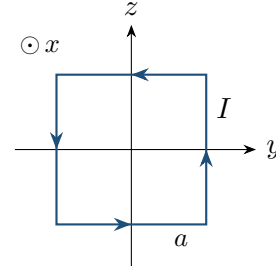
Fields at the points (each wire contributes $\mu_0 I / 2\pi s$; P_1 is at d from wire 1 and $2d$ from wire 2, P_2 at $d/2$ from each, P_3 at $2d$ from wire 1 and d from wire 2):

Point	(i) Parallel currents	(ii) Antiparallel currents
P_1	$\frac{\mu_0}{2\pi d} \left(I_1 + \frac{I_2}{2} \right)$	$\frac{\mu_0}{2\pi d} \left(I_1 - \frac{I_2}{2} \right)$
P_2	$\frac{\mu_0}{\pi d} I_1 - I_2 $	$\frac{\mu_0}{\pi d} (I_1 + I_2)$
P_3	$\frac{\mu_0}{2\pi d} \left(I_2 + \frac{I_1}{2} \right)$	$\frac{\mu_0}{2\pi d} \left(I_2 - \frac{I_1}{2} \right)$

(The directions follow from the right-hand rule: into/out of the page.)

Problem 5 A field $\mathbf{B} = kz\hat{\mathbf{x}}$ (k constant). Find the force on a square loop of side a in the yz -plane, centred at the origin, carrying current I counter-clockwise when viewed down the x -axis. $\mathbf{F} = I \oint d\mathbf{l} \times \mathbf{B}$.

- Top side ($z = a/2$): $B = \frac{ka}{2}\hat{\mathbf{x}}$, $\mathbf{F}_{\text{top}} = \frac{Ika^2}{2}\hat{\mathbf{z}}$.
- Bottom side ($z = -a/2$): $B = \frac{ka}{2}(-\hat{\mathbf{x}})$ and current reversed, $\mathbf{F}_{\text{bottom}} = \frac{Ika^2}{2}\hat{\mathbf{z}}$.
- Vertical sides: $\mathbf{B}$ changes sign between the upper and lower halves, so the forces cancel: $\mathbf{F}_{\text{left}} = \mathbf{F}_{\text{right}} = 0$.



$$\mathbf{F} = Ika^2\hat{\mathbf{z}}$$

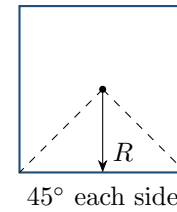
Problem 6 (a) Field at the centre of a square loop carrying current I , where R is the distance from the centre to each side. (b) Field at the centre of a regular n -sided polygon; check $n \rightarrow \infty$.

(a) Each side is a finite wire at distance R subtending $\theta_1 = -45^\circ$, $\theta_2 = +45^\circ$:

$$B_{\text{side}} = \frac{\mu_0 I}{4\pi R} [\sin 45^\circ - (-\sin 45^\circ)] = \frac{\sqrt{2}\mu_0 I}{4\pi R}$$

All four sides contribute in the same direction:

$$B = 4B_{\text{side}} = \frac{\sqrt{2}\mu_0 I}{\pi R}$$



(b) Each side now subtends half-angle π/n :

$$B = \frac{n\mu_0 I}{2\pi R} \sin\left(\frac{\pi}{n}\right)$$

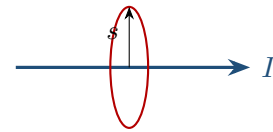
Check $n = 4$: $B = \frac{4\mu_0 I}{2\pi R} \cdot \frac{1}{\sqrt{2}} = \frac{\sqrt{2}\mu_0 I}{\pi R} \checkmark$. A circle is the limit $n \rightarrow \infty$, where $\sin(\pi/n) \rightarrow \pi/n$:

$$B = \frac{\mu_0 I}{2R}$$

Problem 7 Field at distance s from a long straight wire (Ampère's law).

Choose a circular Amperian loop of radius s around the wire, so $I_{\text{enc}} = I$ and $\mathbf{B}$ is tangent with constant magnitude:

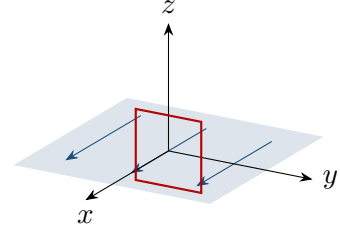
$$\oint \mathbf{B} \cdot d\mathbf{l} = B \cdot 2\pi s = \mu_0 I \quad \Rightarrow \quad \mathbf{B} = \frac{\mu_0 I}{2\pi s} \hat{\phi}$$



Problem 8 Field of an infinite uniform surface current $\mathbf{K} = K\hat{\mathbf{x}}$ flowing over the xy -plane.

By symmetry $\mathbf{B}$ is along $\mp\hat{\mathbf{y}}$ above/below the plane. Take a rectangular Amperian loop in the yz -plane of width l , straddling the sheet. The vertical sides contribute nothing; the horizontal sides each give Bl :

$$2Bl = \mu_0 I_{\text{enc}} = \mu_0 Kl \quad \Rightarrow \quad B = \frac{\mu_0 K}{2}$$



$$\mathbf{B} = \begin{cases} -\frac{\mu_0 K}{2} \hat{\mathbf{y}}, & z > 0 \\ +\frac{\mu_0 K}{2} \hat{\mathbf{y}}, & z < 0 \end{cases}$$

The field is independent of distance from the sheet. This is the analogue of an infinite charged plane: replacing $\mu_0 \rightarrow 1/\epsilon_0$ and $K \rightarrow \sigma$ gives $\mathbf{E} = \pm \frac{\sigma}{2\epsilon_0} \hat{\mathbf{z}}$.

Problem 9 Vector potential of an infinite solenoid (n turns per unit length, radius R , current I). Since the current extends to infinity we can't use the integral formula for $\mathbf{A}$. Instead use the flux:

$$\Phi = \int \mathbf{B} \cdot d\mathbf{a} = \int (\nabla \times \mathbf{A}) \cdot d\mathbf{a} = \oint \mathbf{A} \cdot d\mathbf{l}$$

Inside the solenoid, $B = \mu_0 nI$ along the axis. For a circular loop of radius s :

Inside ($s < R$): $\mu_0 nI \pi s^2 = A \cdot 2\pi s$

$$\mathbf{A}_{\text{in}} = \frac{\mu_0 nI}{2} s \hat{\phi}$$

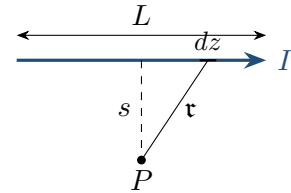
Outside ($s > R$): the flux is only through the cross-section, $\mu_0 nI \pi R^2 = A \cdot 2\pi s$

$$\mathbf{A}_{\text{out}} = \frac{\mu_0 nIR^2}{2s} \hat{\phi}$$

Problem 10 Vector potential of a finite straight wire of length L carrying current I , at perpendicular distance s from its midpoint.

$$\mathbf{A} = \frac{\mu_0}{4\pi} \int \frac{I dz}{\mathbf{r}} = \frac{\mu_0 I}{4\pi} \int_{-L/2}^{L/2} \frac{dz}{\sqrt{s^2 + z^2}} \hat{\mathbf{z}}$$

$$\mathbf{A} = \frac{\mu_0 I}{4\pi} \ln \left[z + \sqrt{z^2 + s^2} \right]_{-L/2}^{L/2} \hat{\mathbf{z}}$$



$$\mathbf{A} = \frac{\mu_0 I}{4\pi} \ln \left[\frac{L/2 + \sqrt{L^2/4 + s^2}}{-L/2 + \sqrt{L^2/4 + s^2}} \right] \hat{\mathbf{z}}$$

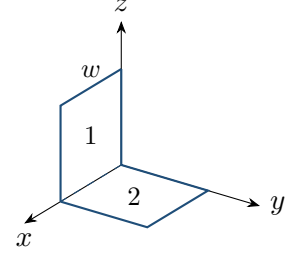
Long wire ($L \gg s$): $\sqrt{L^2/4 + s^2} \approx \frac{L}{2} \left(1 + \frac{2s^2}{L^2} \right)$, so the bracket $\approx \frac{L + s^2/L}{s^2/L} = \frac{L^2}{s^2} + 1 \approx \left(\frac{L}{s} \right)^2$:

$$\mathbf{A} \approx \frac{\mu_0 I}{2\pi} \ln \left(\frac{L}{s} \right) \hat{\mathbf{z}}$$

Problem 11 Magnetic dipole moment of a “bookend”-shaped loop: all sides length w , current I . Split it into two square loops sharing the common edge (the currents on that edge cancel):

- Loop 1 lies in the xz -plane: $\mathbf{m}_1 = Iw^2 \hat{\mathbf{y}}$
- Loop 2 lies in the xy -plane: $\mathbf{m}_2 = Iw^2 \hat{\mathbf{z}}$

$$\mathbf{m} = \mathbf{m}_1 + \mathbf{m}_2 = Iw^2(\hat{\mathbf{y}} + \hat{\mathbf{z}}), \quad |\mathbf{m}| = \sqrt{2} Iw^2$$



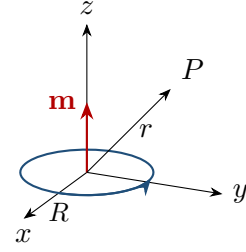
Problem 12 A circular loop of radius R in the xy -plane, centred at the origin, carries current I counter-clockwise (viewed from $+z$). (a) Find its magnetic dipole moment. (b) Find the approximate field far away. (c) Find $\mathbf{A}$ on the axis.

(a) Far away the loop acts like a single point dipole:

$$\mathbf{m} = I\mathbf{a} = I\pi R^2 \hat{\mathbf{z}}$$

(b) The dipole field at (r, θ) :

$$\mathbf{B}_{\text{dip}} = \frac{\mu_0 m}{4\pi r^3} (2 \cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\boldsymbol{\theta}})$$



$$\mathbf{B} \approx \frac{\mu_0 I R^2}{4r^3} (2 \cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\boldsymbol{\theta}})$$

(c) The dipole vector potential is $\mathbf{A}_{\text{dip}} = \frac{\mu_0}{4\pi} \frac{m \sin \theta}{r^2} \hat{\boldsymbol{\phi}}$. On the axis $\theta = 0$, so

$$\mathbf{A}_{\text{axis}} = 0$$